

SINTESI RETI LOGICHE COMBINATORIE

ESERCIZIO 1 - SLIDES

1

X_3	X_2	X_1	X_0	Z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	-
1	0	1	1	-
1	1	0	0	-
1	1	0	1	-
1	1	1	0	-
1	1	1	1	-

DATO UN NUMERO BCD A 4 BIT, PROGETTARE UNA RETE CHE INDICHI SE È MAGGIORE DI 5.

$Z \backslash X_3 X_2$	00	01	11	10
00	0	0	0	0
01	0	0	1	1
11	-	-	-	-
10	1	1	-	-

$$Z_{POS} = (X_2 + X_3)(X_3 + X_4)$$

$$Z_{SOP} = X_3 + X_1 X_2$$

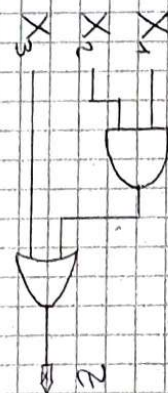
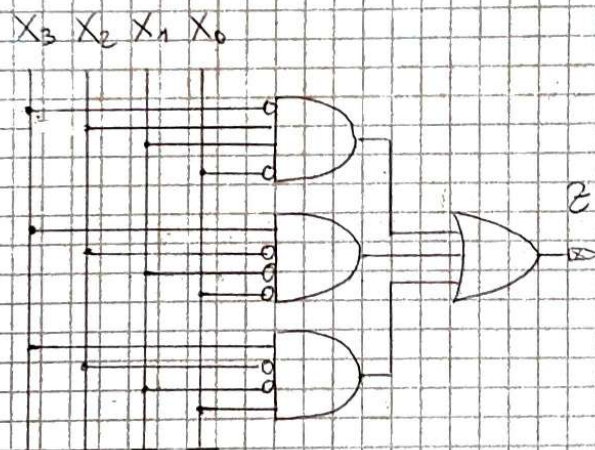
FORME CANONICHE EQUIVALENTI

$$Z_{CAN} = (X_3 + X_2 + X_1 + X_0)(X_3 + X_2 + X_1 + \bar{X}_0)(X_3 + X_2 + \bar{X}_1 + X_0) \cdot (X_3 + \bar{X}_2 + X_1 + \bar{X}_0) \cdot \dots$$

$$Z_{CSOP} = \bar{X}_3 X_2 X_1 \bar{X}_0 + X_3 \bar{X}_2 \bar{X}_1 \bar{X}_0 + X_3 \bar{X}_2 \bar{X}_1 X_0$$

RETE COMPLETA CANONICA SOP

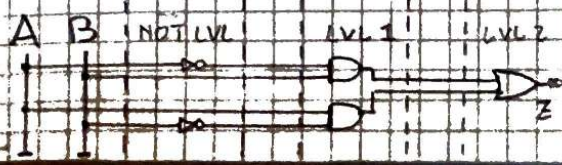
RETE MINIMIZZATA SOP



REGOLE SINTESI RETI LOGICHE COMBINATORIE

- ASSEGNA LE VARIABILI IN ORDINE DECRESCENTE NELLA TABELLA DI VERITÀ: $X_3 X_2 X_1 X_0$, MENTRE NELLE K-MAP RISPETTA I SEGUENTI ORDINI: $X_1 \backslash X_3$ E $X_1 \backslash X_2$ A SECONDA DEL NUMERO DI VARIABILI; QUESTO SEMPLIFICA LO RIEMP.
- IN QUESTO MODO I DUE BIT SULL'ASSE DELLE ORDINATE COINCIDONO CON LE PRIME DUE COLONNE DELLA T.V.
- NELLE K-MAP PIÙ L'AREA COPERTA DAL RAGGRUPPAMENTO È ESTESA, PIÙ VARIABILI VERRANNO OMESSE NELL'EXP. L'OVERLAP DI RAGGRUPPAMENTI È OK, MA NON BISOGNA CREARE RAGGRUPPAMENTI INUTILI. (PREFERISCI K-MAP VERT.).
- PER DECRETARE QUALE TRA LE FORME SOP E POS SIA LA PIÙ EFFICIENTE, DEVO GUARDARE:
 - IL NUMERO COMPLESSIVO DI GATES;
 - IL FAN IN DEI GATE CHE CONNEGGONO LA RETE, SIA AL LVL 1 SIA AL LVL 2.

MIGLIOR MODO PER DISEGNARE LE RETI LOGICHE: CONVIENE IMPARARE LE NOTAZ. DELLE COMPONENTI LOGICHE:



X		
Y	1 BIT	
ON	F.A.	

Z
CAN

ESERCIZIO 2 - SLIDES

X_1	X_0	Y_1	Y_0	G	E	L
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

PROGETTARE UN COMPARATORE A 2 BIT.

G

$X_1 \backslash X_0$	00	01	11	10
00	0	0	0	0
01	1	0	0	0
11	1	1	0	1
10	1	1	0	0

L

$X_1 \backslash X_0$	00	01	11	10
00	0	1	1	1
01	0	0	1	1
11	0	0	0	0
10	0	0	0	1

$$G_{SOP} = X_1 \bar{Y}_1 + X_1 X_0 \bar{Y}_0 + X_0 \bar{Y}_1 \bar{Y}_0 \quad L_{SOP} = \bar{X}_1 \bar{X}_0 Y_0 + \bar{X}_0 Y_1 \bar{Y}_0 + X_1 Y_1$$

E

$X_1 \backslash X_0$	00	01	11	10
00	1	0	0	0
01	0	1	0	0
11	0	0	1	0
10	0	0	0	1

FAN IN LIVEL 1 = 4

FAN IN LIVEL 2 = 4

FAN IN LIVEL 1 = 2 MENO ENTRATE!

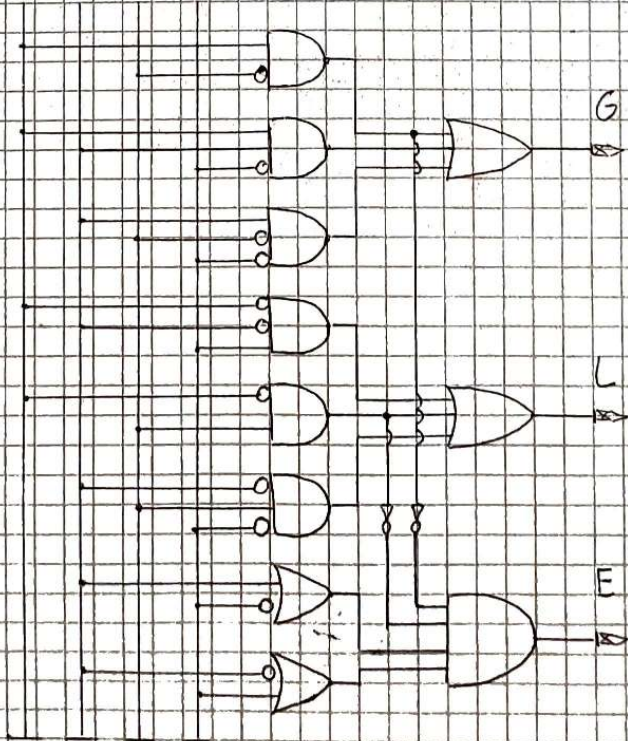
FAN IN LIVEL 2 = 4

$$E_{POS} = (X_0 + \bar{Y}_0)(\bar{X}_0 + Y_0)(X_1 + \bar{Y}_1)(\bar{X}_1 + Y_1)$$

$$E_{SOP} = \bar{X}_1 \bar{X}_0 \bar{Y}_1 \bar{Y}_0 + \bar{X}_1 X_0 \bar{Y}_1 Y_0 + X_1 X_0 Y_1 Y_0 + X_1 \bar{X}_0 Y_1 \bar{Y}_0$$

RETE MINIMIZZATA

$X_1 X_0 Y_1 Y_0$



SINTESI RETI LOGICHE COMBINATORIE

2

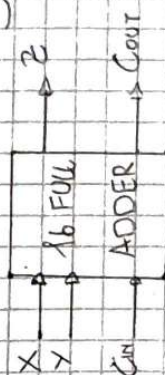
ESERCIZIO 3 - SLIDES (SONNATORE AD 1 BIT, CON CARRY IN)

X	Y	C _{IN}	Z	C _{OUT}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Z	C _{IN}	0	1
X \ Y			
00	0	1	
01	1	0	
11	0	1	
10	1	0	

C _{OUT}	C _{IN}	0	1
X \ Y			
00	0	0	
01	0	1	
11	1	1	
10	0	1	

COMPONENTE FA



$$Z_{SOP} = \overline{X}\overline{Y}C_{IN} + \overline{X}Y\overline{C_{IN}} + XY\overline{C_{IN}} + XYC_{IN}$$

$$C_{OUT}SOP = YC_{IN} + XY + XC_{IN}$$

ESERCIZIO 4 - SLIDES (SONNATORE AD 2 BIT, SENZA CARRY IN)

X ₁	X ₀	Y ₁	Y ₀	S ₂	S ₁	S ₀
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0

S ₂	Y ₁ Y ₀	00	01	11	10
X ₁ X ₀					
00	0	0	0	0	0
01	0	0	1	0	
11	0	1	1	1	
10	0	0	1	1	

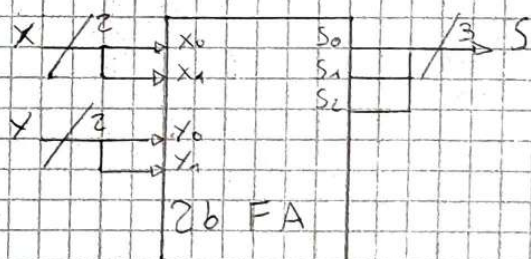
S ₁	Y ₁ Y ₀	00	01	11	10
X ₁ X ₀					
00	0	0	1	1	
01	0	1	0	1	
11	1	0	1	0	
10	1	1	0	0	

$$S_2SOP = X_1X_1 + X_0Y_1Y_0 + X_1X_0Y_0$$

$$S_1SOP = \overline{X_1}\overline{X_0}Y_1 + \overline{X_1}Y_1\overline{Y_0} + \overline{X_1}X_0\overline{Y_1}Y_0 + X_1X_0Y_1Y_0 + X_1\overline{Y_1}\overline{Y_0} + X_1\overline{X_0}\overline{Y_1}$$

S ₀	Y ₁	00	01	11	10
X ₁ X ₀					
00	0	1	1	0	
01	1	0	0	1	
11	1	0	0	1	
10	0	1	1	0	

$$S_0SOP = \overline{X_0}Y_0 + X_0\overline{Y_0} \rightarrow X_0 \oplus Y_0$$



ESERCIZIO 5 SLIDES (SOMMATTORE AD 2 BIT, CON CARRY IN.)

X_1	X_0	Y_1	Y_0	C_{in}	S_2	S_1	S_0
0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	1
0	0	0	1	0	0	0	1
0	0	0	1	1	0	1	0
0	0	1	0	0	0	1	0
0	0	1	0	1	0	1	1
0	0	1	1	0	0	1	1
0	0	1	1	1	1	0	0
0	1	0	0	0	0	0	1
0	1	0	0	1	0	1	0
0	1	0	1	0	0	1	0
0	1	0	1	1	0	1	1
0	1	1	0	0	0	1	1
0	1	1	0	1	1	0	0
0	1	1	1	0	1	0	0
0	1	1	1	1	1	0	1
1	0	0	0	0	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	0	0	1	1
1	0	0	1	1	1	0	0
1	0	1	0	0	1	0	0
1	0	1	0	1	1	0	1
1	0	1	1	0	1	0	1
1	0	1	1	1	1	0	1
1	1	0	0	0	1	0	1
1	1	0	0	1	1	0	1
1	1	0	1	0	1	1	0
1	1	0	1	1	1	1	0
1	1	1	0	0	1	1	0
1	1	1	0	1	1	1	0
1	1	1	1	0	1	1	0
1	1	1	1	1	1	1	1

MAPPE VARIABLE S_2

$C_{in} = 0$

S_2	Y_1	Y_0	X_1	X_0	00	01	11	10
00	0	0	0	0	0	0	0	0
01	0	0	0	1	0	0	1	0
11	0	1	1	1	0	1	1	1
10	0	0	1	1	0	0	1	1

$C_{in} = 1$

S_2	Y_1	Y_0	X_1	X_0	00	01	11	10
00	0	0	0	0	0	0	1	0
01	0	0	0	1	0	0	1	1
11	1	1	1	1	1	1	1	1
10	0	1	1	1	0	1	1	1

$$S_2 S_0 P = X_1 Y_1 + X_0 Y_0 Y_1 + X_1 X_0 Y_0$$

$$S_2 S_0 P = Y_1 Y_0 + X_1 X_0 + X_0 Y_1 + X_1 Y_0 + X_1 Y_1$$

MAPPE VARIABLE S_1

$C_{in} = 0$

S_1	Y_1	Y_0	X_1	X_0	00	01	11	10
00	0	0	0	0	0	0	1	1
01	0	1	0	1	0	1	0	1
11	1	0	1	0	1	0	1	0
10	1	1	1	0	1	1	0	0

$C_{in} = 1$

S_1	Y_1	Y_0	X_1	X_0	00	01	11	10
00	0	1	0	0	0	1	0	1
01	1	1	0	1	1	1	0	0
11	0	0	1	1	0	0	1	1
10	1	0	1	1	1	0	1	0

$$S_1 S_0 P = \bar{X}_1 \bar{X}_0 Y_1 + \bar{X}_1 X_1 \bar{Y}_0 + \bar{X}_1 X_0 \bar{Y}_1 Y_0 + X_1 X_0 Y_1 Y_0 + X_1 X_0 \bar{Y}_1 + \bar{X}_1 \bar{Y}_1 Y_0 + \bar{X}_1 X_0 \bar{Y}_1 + X_1 X_0 Y_1 + X_1 \bar{Y}_0 \bar{Y}_1$$

$$S_1 S_0 P = \bar{X}_1 \bar{X}_0 Y_1 \bar{Y}_0 + X_1 \bar{X}_0 \bar{Y}_1 \bar{Y}_0$$

MAPPE VARIABLE S_0

$C_{in} = 0$

S_0	Y_1	Y_0	X_1	X_0	00	01	11	10
00	0	1	1	0	0	1	1	0
01	1	0	0	1	1	0	0	1
11	1	0	0	1	1	0	0	1
10	0	1	1	0	0	1	1	0

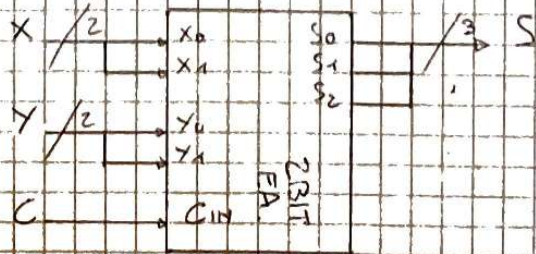
$C_{in} = 1$

S_0	Y_1	Y_0	X_1	X_0	00	01	11	10
00	1	0	0	0	1	0	0	1
01	0	1	1	0	0	1	1	0
11	0	1	1	0	0	1	1	0
10	1	0	0	0	1	0	0	1

$$S_0 S_0 P = \bar{X}_0 X_0 + X_0 \bar{Y}_0$$

$$S_0 S_0 P = \bar{X}_0 \bar{Y}_0 + X_0 Y_0$$

COMPONENTE



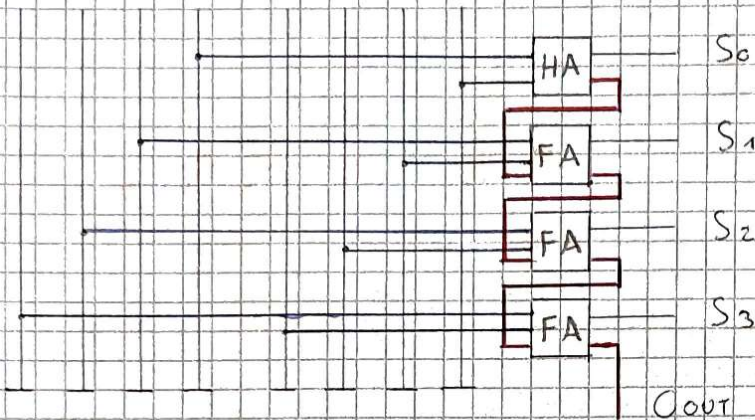
SINTESI RETI LOGICHE COMBINATORIE

3

PROGETTARE UN SOMMATORE USANDO HALF E FULL ADDER

SOMMATORE MODULARE A 4 BIT

$X_3 X_2 X_1 X_0 \quad Y_3 Y_2 Y_1 Y_0$



ESERCIZIO 6 - SLIDES

A \ B	00	01	11	10
00	0	0	1	1
01	0	1	1	1
11	0	1	1	1
10	1	1	1	1

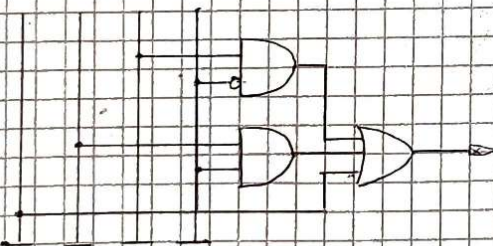
$$SOP_{CANONICA} = \bar{C}\bar{D}A\bar{B} + \bar{C}\bar{D}A\bar{B} + \bar{C}\bar{D}A\bar{B} + \bar{C}\bar{D}A\bar{B} + \bar{C}\bar{D}A\bar{B} + C\bar{D}A\bar{B} + C\bar{D}A\bar{B} + C\bar{D}A\bar{B} + C\bar{D}A\bar{B} + C\bar{D}A\bar{B}$$

$$SOP_{MINIMA} = A + DB + CB$$

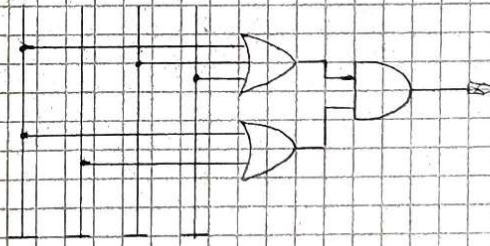
$$POS_{CANONICA} = (C+D+A+B)(C+D+A+\bar{B})(C+\bar{D}+A+B)(\bar{C}+\bar{D}+A+B)$$

$$POS_{MINIMA} = (A+B)(C+D+A)$$

A B C D



A B C D



ESERCIZIO 7 - SLIDES

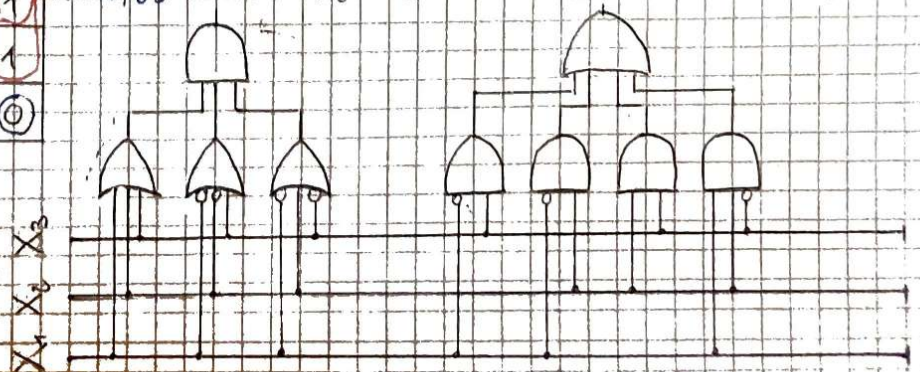
X_1	X_2	X_3	L
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$X_1 X_2 X_3$

X_1	X_2	X_3
0	0	1
1	0	1
0	1	1
1	1	0

$$LSOP = \bar{X}_1 X_3 + \bar{X}_1 X_2 + X_2 X_3 + X_1 X_2 \bar{X}_3$$

$$LPOS = (X_1 + X_2 + X_3)(\bar{X}_1 + \bar{X}_2 + X_3)(\bar{X}_1 + X_3 + \bar{X}_2)$$



ESERCIZIO 8 - SLIDES

IR_1	IR_2	IR_3	IS_1	IS_2	IS_3
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	0
1	0	0	1	0	0
1	0	1	1	0	0
1	1	0	1	0	0
1	1	1	1	0	0

S_1	R_3	
IR_1	IR_2	
00	0	1
01	0	0
11	1	1
10	1	1

S_2	R_3	
IR_1	IR_2	
00	0	0
01	1	1
11	0	0
10	0	0

S_3	R_3	
IR_1	IR_2	
00	0	1
01	0	0
11	0	0
10	0	0

$$IS_1 = IR_1$$

$$IS_2 = \overline{IR_1} IR_2$$

$$IS_3 = \overline{IR_1} \overline{IR_2} IR_3$$